

OpenMind: Query-Time Diversification via Logical Rules

This survey only takes you less than 15 minutes to complete.

IMPORTANT: Please read the instruction carefully and select your answers. We will check the consistency of your answers to make sure that you do not answer the questions by random selection.

CAUTION: If you provide inconsistent answers, your result will get **REJECTED!**

* Indicates required question

Preliminary

We will consider an anime recommendation scenario, which recommends some anime to users.

We aim to provide diversified recommendations for user to break information cocoon caused by the traditional recommender system.

We will give the user profiles and some historical interactions of a user u_0 .

Based on such information, we will show you FOUR recommended items to user u_0 such that user u_0 can be exposed to the anime with different viewpoints/perspectives which can be easily accepted by user u_0 , yielding to break information cocoon caused by the traditional recommender system.

User Profiles

We provide the user profile in the below, including his favorite anime.

User Name	Gender	Birthday	Age Group
Visceras	Male	Aug 18, 1997	Post-95s

His Favorite Anime

Below are his favorite animes.

Each anime is provided some information, including the year, the genres and the description.

Favorite Anime	Year	Genre	Anime Description
Clannad: After Story	2008	Drama/ romance/ slice of life/ supernatural	A sequel to Clannad that focuses on adulthood, family, loss, and emotional growth.
Fate/Zero	2011	Action/ dark fantasy/ supernatural	A battle-royale story in which mages and heroic spirits fight in the Holy Grail War.
Death Note	2006	Psychological thriller/ mystery/ supernatural	A high school student gains a notebook that can kill anyone whose name is written in it.
Wolf Children	2012	Drama/ fantasy/ family/ slice of life	A heartfelt film about a mother raising two half-wolf children while protecting their secret.



Clannad: After Story



Fate/Zero



Death Note



Wolf Children

His Latest Watched List

We provide his latest watched list in below.

Each anime is provided some information, including the year, the genres and the description.

Watched Anime	Year	Genre	Anime Description
Terror in Resonance	2014	Psychological thriller/ mystery/ crime	Two teenage terrorists challenge society through a series of attacks and riddles.
Phantom: Requiem for the Phantom	2009	Psychological/ crime/ action/ drama	A young man is drawn into the underworld after being turned into an assassin.
Tokyo Ghoul	2014	Dark fantasy/ horror/ action/ supernatural	A college student becomes half-ghoul and struggles between human and monster worlds.
Fate/Zero	2011	Action/ dark fantasy/ supernatural	A battle-royale story in which mages and heroic spirits fight in the Holy Grail War.



Terror in Resonance



Phantom: Requiem
for the Phantom



Tokyo Ghoul



Fate/Zero

Short User Preference

Overall, this user appears to prefer anime with **dark tone, psychological tension, suspense, crime-related plots, supernatural elements, and action-oriented narratives.**

Traditional Recommendation

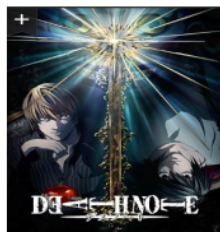
Based on the user profile, his favorite anime and his latest watched anime, a traditional recommender system recommends him an anime, **Death Note**, as shown in the below.

Traditional Recommendation- Death Note

Its genres consist of **psychological thriller, mystery, supernatural**.

Short description:

A high school student gains a notebook that can kill anyone whose name is written in it.

	Anime	Year	Genre	Anime Description
	Death Note	2006	Psychological thriller/ mystery/ supernatural	A high school student gains a notebook that can kill anyone whose name is written in it.

Objective

However, such recommendation (**Death Note**) keeps the user in the information cocoon.

Because it **strongly overlaps with the user's historical preferences for dark, psychological, suspenseful, crime-related, and supernatural content**. Rather than broadening the user's exposure, such a recommendation mainly reinforces an already dominant preference pattern.

Based on the recommended anime (Death Note), we would like to provide diversified recommendations for this user to break this information cocoon.

The diversified recommended item should be easily accepted by user and at the same time, provide alternative viewpoints/perspectives with respect to the traditionally recommended anime (Death Note).

Questions

Consider the following **four recommendations** for making the user *u0* exposed to diverse items to broaden his views or perspectives.

You will be asked **eight questions** based on the four recommendations.

Recommendation 1: Howl's Moving Castle

Its genres consist of fantasy, adventure and romance.

Short description:

A young woman cursed into an old body becomes involved with a mysterious wizard and his moving castle

	Anime	Year	Genre	Anime Description
	Howl's Moving Castle	2004	Fantasy/ adventure/ romance	A young woman cursed into an old body becomes involved with a mysterious wizard and his moving castle.

Recommendation 2: The Future Diary

Its genres consist of psychological thriller, survival, supernatural and action.

Short description:

A boy is forced into a deadly game where participants use diaries that can predict the future.

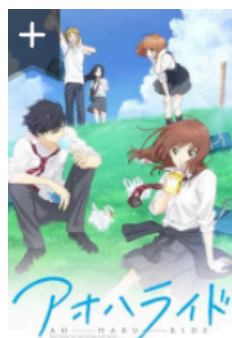
	Anime	Year	Genre	Anime Description
	The Future Diary	2011	Psychological thriller/ survival/ supernatural/ action	A boy is forced into a deadly game where participants use diaries that can predict the future

Recommendation 3: Blue Spring Ride

Its genres consist of romance, drama, slice of life and school.

Short description:

A high school girl reconnects with her first love and navigates youth, friendship, and emotional change.



Anime	Year	Genre	Anime Description
Blue Spring Ride	2014	Romance/ drama/ slice of life/ school	A high school girl reconnects with her first love and navigates youth, friendship, and emotional change.

Recommendation 4: Death Note: Rewrite

Its genres consist of psychological thriller, mystery and supernatural.

Short description:

A recap film that reorganizes the main events of *Death Note* into a condensed retelling.



Anime	Year	Genre	Anime Description
Death Note: Rewrite	2007	Psychological thriller/ mystery/ supernatural	A recap film that reorganizes the main events of Death Note into a condensed retelling.

Evaluation

Consider the quality of the above four recommendations for breaking information cocoon. In particular, focus on the following four aspects of each recommendation with respect to the traditional recommendation (*namely*, **Death Note**):

1. Acceptability
2. Topic Consistency
3. View Divergence
4. Overall Performance

Acceptability

"Acceptability" is an aspect to be evaluated in this survey.

A given diversified recommendation is considered as good in terms of acceptability **if it can be easily accepted by the user u_0** .

Thus, a recommendation with higher acceptability is better.

1. Q1. Consider only "**Acceptability**" of each recommendation. Please indicate one recommendation that you think with the best "Acceptability" in all four diversified recommendations. *

Suppose that you are user u_0 .

You could make decision **by considering the user profile, favorite anime and latest watched anime**.

Mark only one oval.

- ☐ Recommendation 1
- ☐ Recommendation 2
- ☐ Recommendation 3
- ☐ Recommendation 4

2. Q2. Consider only "Acceptability" of each recommendation. Please indicate ***exactly 2 recommendations*** that you think with better "Acceptability" in all four recommendations. Note: One of them ***must*** be the recommendation you indicated in Q1. *

Check all that apply.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

Topic Consistency

"Topic Consistency" is an aspect to be evaluated in this survey.

A given recommendation is considered as good in terms of topic consistency **if it maintains consistent topic with the traditional recommendation (Death Note).**

Thus, a recommendation with higher consistency is better.

3. Q3. Consider only **"Topic Consistency"** of each recommendation. Please indicate one recommendation that you think with the best "Topic Consistency". *

Mark only one oval.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

4. Q4. Consider only **"Topic Consistency"** of each recommendation. Please indicate **exactly 2 recommendations** that you think with better "Topic Consistency" in all four recommendations. Note: One of them **must** be the recommendation you indicated in Q3. *

Check all that apply.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

View Divergence

"View Divergence" is an aspect to be evaluated in this survey.

View Divergence measures whether the recommendation provides a **different perspective or worldview** from the reference work, rather than simply a different genre or topic.

A high-scoring recommendation should differ in **moral stance, narrative framing, character perspective, or interpretation of shared core issues**, while remaining comparable at the content level.

Therefore, **genre/theme difference alone should not be considered sufficient** for view divergence.

Thus, a recommendation that provides different viewpoint/perspectives is better than one that falls the same flow of narratives.

5. Q5. Consider only "**View Divergence**" of each recommendation. Please indicate * one recommendation that you think with the best "View Divergence".

Mark only one oval.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

6. Q6. Consider only "**View Divergence**" of each recommendation. Please indicate * ***exactly 2 recommendations*** that you think with better "View Divergence" in all four recommendations. Note: One of them ***must*** be the recommendation you indicated in Q5.

Check all that apply.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

Overall Performance

"Overall Performance" is the last aspect to be evaluated in this survey.

It is an aspect of the overall performance of each recommendation.

Under the aspect of the overall performance,

we would like to have **a recommendation such that it is good in terms of the acceptability, topic consistency and view divergency simultaneously.**

In this case, the user $u0$ can accept this recommendation with high probability, broadening his horizons,

and then, breaking the information cocoon caused by the traditional recommender system.

Thus, if the performance of a recommendation is better than another one in terms of the acceptability, topic consistency and view divergency, the former recommendation is better.

7. Q7. Consider only "Overall Performance" of each recommendation. Please indicate one recommendation that you think with the best "Overall Performance". *

Mark only one oval.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

8. Q8. Consider only "Overall Performance" of each recommendation. Please indicate **exactly 2 recommendations** that you think with better "Overall Performance" in all four recommendations. Note: One of them **must** be the recommendation you indicated in Q7. *

Check all that apply.

- ☐ recommendation 1
- ☐ recommendation 2
- ☐ recommendation 3
- ☐ recommendation 4

You are almost done!

We will use your nickname to verify your work.
Please enter your nickname below.

9. **Nickname**

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